

VILSANDI, Estonia

WMO: 262140

Lat: 58.3828N

Lon: 21.8142E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.4	-10.3	-16.4	0.9	-12.5	-13.2	1.2	-9.3	16.3	0.7	14.8	0.6	4.1	80	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	25.5	20.0	23.7	19.3	22.1	18.4	21.0	24.1	19.9	22.9	18.9	21.6	4.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.4	23.0	18.7	13.5	21.9	17.7	12.7	20.8	60.5	24.3	57.0	23.0	53.6	21.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 14.4	12.7	11.4		-14.5	27.4	4.5	2.4	-17.7	29.1	-20.3	30.5	-22.8	31.8	-26.1		
(5)				-15.1	21.8	4.2	1.4	-18.1	22.8	-20.5	23.7	-22.9	24.4	-25.9	25.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-1.2	-1.7	0.6	4.7	10.3	14.3	18.2	18.2	14.0	8.7	4.8	2.3
	DBStd	7.69	4.53	4.46	3.09	2.76	3.62	2.78	2.67	2.34	2.57	3.29	3.21	3.93
	HDD10.0	1631	346	328	292	162	38	1	0	0	4	63	156	240
	HDD18.3	3898	604	562	551	409	251	125	34	29	130	298	406	498
	CDD10.0	838	0	0	0	3	47	131	254	255	124	23	0	0
	CDD18.3	64	0	0	0	0	2	5	31	26	1	0	0	0
	CDH23.3	195	0	0	0	0	7	22	101	66	0	0	0	0
	CDH26.7	22	0	0	0	0	0	2	12	8	0	0	0	0
Wind	WSAvg	5.6	6.4	5.8	5.6	5.3	4.7	4.9	4.6	5.0	5.3	6.1	6.3	7.3
Precipitation	PrecAvg	581	45	34	32	28	32	50	52	74	58	64	64	51
	PrecMax	796	113	70	68	59	63	96	105	147	118	110	114	115
	PrecMin	398	9	2	9	6	7	6	3	2	11	9	4	19
	PrecStd	100	22	16	15	16	15	21	27	40	29	30	28	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	4.7	9.6	16.6	24.2	26.2	28.2	27.6	21.2	15.5	10.9	8.9
		MCWB	5.9	3.7	5.7	9.9	16.0	18.3	21.1	21.1	17.3	13.8	10.0	7.9
	2%	DB	5.8	4.1	6.2	13.1	20.9	22.3	25.8	25.0	19.3	14.6	10.1	8.1
		MCWB	5.1	3.3	4.1	8.1	15.3	17.6	20.6	20.1	16.1	13.0	9.1	7.1
	5%	DB	4.9	3.6	4.7	10.5	18.2	19.8	24.2	23.3	18.2	13.8	9.4	7.3
		MCWB	3.9	2.9	3.4	6.8	14.0	16.1	20.1	19.3	15.5	12.4	8.3	6.3
	10%	DB	4.1	3.0	3.9	8.7	15.8	18.2	22.5	21.7	17.4	13.0	8.7	6.4
		MCWB	3.2	2.5	2.9	6.1	12.6	15.4	19.0	18.3	15.1	11.7	7.6	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	4.1	6.2	10.2	17.7	19.8	22.6	22.1	18.2	14.4	10.2	8.3
		MCDB	6.4	4.4	8.8	15.2	22.2	24.4	26.1	26.0	20.4	15.0	10.6	8.7
	2%	WB	5.3	3.5	4.5	8.5	16.1	17.9	21.4	20.7	17.0	13.2	9.3	7.3
		MCDB	5.7	3.9	5.9	12.1	19.6	21.6	24.6	23.7	18.6	14.1	9.9	8.0
	5%	WB	4.1	3.0	3.7	7.3	14.4	16.7	20.4	19.8	16.1	12.6	8.5	6.4
		MCDB	4.7	3.5	4.5	9.7	17.6	19.4	23.4	22.6	17.8	13.6	9.2	7.2
	10%	WB	3.3	2.5	3.1	6.2	12.6	15.7	19.3	18.8	15.3	11.9	7.8	5.6
		MCDB	4.0	3.0	3.7	8.2	15.7	17.8	22.2	21.2	17.0	12.9	8.5	6.3
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.4	3.5	4.4	5.2	4.4	4.5	4.5	3.6	3.2	2.6	2.6
		MCDBR	2.4	2.2	4.3	7.9	9.3	7.7	7.2	6.9	4.7	3.2	2.3	2.1
		MCWBR	2.7	2.1	3.1	4.3	5.2	4.1	3.6	3.7	3.2	2.9	2.5	2.2
	5% WB	MCDBR	2.4	2.1	3.9	6.6	8.5	6.6	6.4	5.9	3.8	3.1	2.2	2.0
		MCWBR	2.7	2.1	3.0	4.1	5.2	3.9	3.6	3.7	3.1	3.0	2.5	2.2
Clear-Sky Solar Irradiance	taub	0.295	0.322	0.341	0.375	0.369	0.367	0.395	0.388	0.357	0.342	0.326	0.288	
	taud	2.213	2.271	2.327	2.328	2.421	2.468	2.409	2.415	2.481	2.403	2.237	2.154	
	Ebn at Noon	529	699	801	831	865	871	833	812	783	675	478	402	
	Edh at Noon	48	73	90	107	104	101	105	97	79	64	47	37	
All-Sky Solar Radiation	RadAvg	0.29	0.86	2.39	4.18	5.79	6.18	5.79	4.55	2.88	1.28	0.38	0.20	
	RadStd	0.04	0.15	0.25	0.45	0.50	0.31	0.45	0.37	0.33	0.16	0.07	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	-115	N/A	N/A	N/A

Nomenclature: See separate page